PICO: A Solo-Operator Framework for Small-Budget Polymarket Bots

Preprint. Under review.

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Abstract

Prediction-market environments like Polymarket impose distinctive frictions when a single operator manages a modest bankroll. We propose PICO, a modular evaluation framework that disentangles market selection, entry/exit logic, sizing, risk controls, cooldowns, paper-trading, and backtesting into a reproducible pipeline tailored for a $100 - -500$ budget. PICO emphasizes observability and a walk-forward evaluation protocol with pre-registered ablations, enabling a solo operator to iterate safely toward robust strategies. Our framing centers on a fixed, small market universe and explicit liquidity-aware decisions, with cost-aware backtesting that accounts for fees, slippage, and settlement delays. The core contribution is a design philosophy and implementation blueprint that makes the evaluation of strategy families and their interactions with liquidity frictions transparent and reproducible. We describe a practical paper-trading setup, a roadmap for translating backtested ideas into live pilots, and concrete artifacts (data splits, seeds, containerized environments, and run logs) to support independent replication. While concrete numerical gains will emerge from future runs, the emphasis is on disciplined baselining, ablations, and observability to identify combinations of market selection criteria, risk limits, and cooldown policies that yield stable growth under tight budgets. The narrative foregrounds practical implications for solo-maintained trading in small prediction-market budgets, with an explicit emphasis on reproducibility and actionable guidance for live pilots.

> **Note:** This paper was produced in degraded mode. Quality gate score (3.0/4.0) was below threshold. Unverified numerical results in tables have been replaced with `--` and require independent verification.

1 Introduction

Prediction markets provide a natural interface for crowdsourced foresight but impose nontrivial frictions when a single operator with a small bankroll participates. Micro-budget traders contend with thin liquidity, wide bid–ask spreads, high relative entry costs, and volatile event horizons that amplify slippage and limit available edge. The design space extends beyond signal quality to encompass disciplined pacing, risk controls, and explicit operational safeguards that keep exposure manageable while preserving learning opportunities. In this context, the motivation for a principled evaluation framework is clear: to separate the edge provided by market understanding from the friction imposed by execution, fees, and liquidity constraints. A framework that makes these tradeoffs explicit allows solo operators to run credible backtests, perform walk-forward validation, and build credible paper-trading setups that translate into real-money practice. The problem is, in brief, how a single trader can credibly evaluate, compare, and iterate viable Polymarket-like trading strategies under liquidity constraints, with transparent artifacts and reproducible results.

Our approach, called PICO, is a modular pipeline designed for a small-budget Polymarket-like context. The Market Selection module identifies markets with favorable liquidity proxies—depth, spread, and proximity to event—while respecting a fixed and small universe. The Entry/Exit module

translates signals into YES/NO bets under liquidity-aware slippage constraints and per-market exposure limits. The Sizing module caps risk at the portfolio level and the per-market level to balance edge capture against diversification. Explicit risk controls implement stop-loss and take-profit semantics, cooldowns, and pacing rules to prevent overtrading. A Paper-Trading and Backtesting engine models costs (fees, slippage) and settlement delays to produce credible PnL trajectories. An Observability subsystem reports per-market liquidity proxies, exposure, cooldown status, and risk alerts. Finally, a Roadmap for robust, solo-maintained strategies emphasizes modular APIs, testability, and reproducibility. The core idea is to provide a disciplined workflow that a solo operator can sustain over months, with pre-registered ablations to isolate the contribution of specific design choices.

Contributions

- A minimal viable evaluation pipeline for micro-budget Polymarket-like trading, including a fixed-universe baseline and pre-registered ablations.
- A clear metrics schema capturing profitability, risk, execution costs, robustness, and observability tailored for solo operators.
- A reproducible artifact package with data partitions, seeds, containerization, and run logs to enable independent replication.
- A practical, stepwise roadmap for scaling from credible baseline to robust, solo-maintained strategies under real-money constraints.

Organization: We first formalize the PICO framework, then detail the evaluation protocol and planned experiments, and finally discuss implications for solo operators and future scaling. Throughout, we foreground citations to relevant literature that supports the feasibility of a market-selection approach in small-budget decision markets, as well as best practices for backtesting, ablations, and reproducibility in AI-enabled trading contexts –[**?**], [**?**], [**?**], [**?**].

2 Related Work

Prediction markets have long been studied for their information-aggregation properties and liquidity challenges. In particular, market design literature emphasizes how liquidity depth, market-making incentives, and event proximity shape price formation and trading costs in Yes/No markets. Unlike macro markets that span long horizons, micro-budget decision markets require careful budgeting, pacing, and risk controls to avoid erosion from churn and illiquidity. Our framing attends to these constraints by explicitly integrating liquidity proxies into market selection and by binding entry decisions to per-market exposure caps. This stands in contrast to broader market design literature that treats liquidity as a macro property rather than a per-market constraint, and it aligns with recent work that identifies the practical importance of cost-aware evaluation pipelines for decision markets , [**?**].

Backtesting and evaluation for AI-enabled trading has highlighted the necessity of realistic costs, data-splitting, walk-forward validation, and robust baselines. Classical finance discussions stress incorporating slippage, fees, and latency, especially when turnover is high and markets are thin . Contemporary AI-enabled trading work extends these concerns to learning-based strategies, stressing that robust pipelines must separate signal quality from execution frictions to avoid overestimating real-world performance [**?**], [**?**], [**?**], [**?**]. In prediction-market contexts, evaluation protocols have lagged behind the advances in finance, creating a gap that PICO seeks to close by bringing principled backtesting, walk-forward validation, and reproducible artifacts into the decision-market domain.

Observability, risk controls, and solo-operator governance are central to practical deployment. The literature highlights dashboards, alerting, and anomaly detection as essential for monitoring exposure, cooldown adherence, and drift in strategy behavior, particularly when a single operator bears full accountability [**?**], [**?**]. The solo-operator lens further motivates per-market and portfolio-level risk controls to prevent runaway turnover in illiquid markets, as governance and trustworthy automation frameworks emphasize clear accountability and auditable decision traces [**?**], [**?**].

Gaps and positioning. While prior work addresses prediction-market dynamics, micro-trading challenges, or AI-assisted evaluation in isolation, there is a lack of a cohesive, reproducible framework tailored to micro-budget, solo-maintained Polymarket-like strategies with explicit liquidity-aware decision policies. PICO positions itself at this intersection by (i) incorporating liquidity-aware market selection, (ii) codifying entry logic for YES/NO bets under tight budgets, (iii) proposing risk-aware sizing with explicit stop-loss and take-profit semantics, (iv) enforcing cooldowns and anti-overtrading mechanisms, (v) embedding a paper-trading and backtesting workflow that models costs realistically, and (vi) delivering observability dashboards and reproducible artifacts for solo operators. This combination aims to make the evaluation of strategy families transparent and extensible, enabling iterative refinement by a single operator.

Notable sources informing PICO's design include: the practical analysis of liquidity, market-making, and event-driven dynamics in prediction markets ; macro and conditional markets as frames for event-driven trading in small venues [**?**], [**?**]; AI-assisted trading frameworks that stress robust evaluation pipelines under realistic costs [**?**], [**?**], [**?**], [**?**]; and empirical work on risk management and constrained trading in crypto and decision markets , [**?**], .

In sum, the Related Work survey motivates a cohesive framework that directly targets micro-budget, solo-maintained Polymarket-like strategies with explicit market selection, exposure control, and rigorous evaluation protocols.

3 Method

Problem formulation and design goals We study a Polymarket-like decision environment in discrete decision epochs t {1,...,T}. Let M(t) denote the set of active YES/NO markets available at time t, and B(t) the trader's bankroll before trades at t. Each market m M(t) exposes two outcomes: YES and NO. A YES bet of size w on market m costs p_m(t) · w and pays out w if the outcome is YES, yielding a profit of w(1p_m(t)) minus Fees. A NO bet of size v costs (1p_m(t)) · v and pays out v if the outcome is NO, with profit v p_m(t) Fees. We model execution costs with three components: per-unit trading fee f, an estimated slippage s_m(t) that scales with stake and liquidity proxies, and a settlement delay g_m(t). The realized PnL for market m, at event resolution, is the payout minus total cost across both sides. The policy objective is to maximize expected, risk-adjusted PnL while respecting liquidity constraints, exposure caps, and pacing rules.

Market selection and liquidity-aware signaling The Market Selection subsystem computes a liquidity-adjusted ranking over M(t) using a feature vector _m(t) that aggregates proxies: bid-ask spread _m(t), depth proxy D_m(t), recent turnover _m(t), time-to-event evt_m(t), and a volatility proxy _m(t). We define a normalized liquidity score L_m(t) [0,1], increasing with depth and decreasing with spreads and time-to-event magnitude, and normalize across markets to identify a top-K subset for potential entry. An action signal F(m,t) {YES, NO, HOLD} is derived from a light predictor or heuristic checks, constrained to a small, interpretable probability range to avoid overfitting in thin markets. The predictor's output and the liquidity signal jointly determine entry eligibility and risk budgets for each market.

Position sizing and risk controls Per-market exposure is bounded by E_m, expressed as a fraction of B(t). The maximum stake on a market is w_m(t) $\leq$ min(E_m B(t), b_max), with b_max a hard cap to prevent over-concentration. The per-trade risk budget assigns a target stake w_m(t) so that the worst-case one-step loss across YES and NO does not exceed a fixed fraction of B(t) ((0,0.1], typically in the 0.02–0.05 range in our planning). The sizing rule enforces w_m(t) $\leq$ max_loss_m / max(p_m(t), 1 p_m(t)), where max_loss_m = B(t). This conservative approach aims to preserve capital in the face of liquidity shocks while preserving the possibility of edge capture in liquid windows.

Exit logic, stop-loss, and take-profit We implement threshold-based stop-loss and take-profit triggers tied to realized PnL since entry. For a position b_m(t0) entered at t0, unrealized PnL depends on the realized outcome when resolved. If PnL_unreal_m(t) breaches a stop-loss _loss, we exit; if it reaches a take-profit _profit, we exit to lock in gains. Thresholds _loss and _profit are calibrated to market liquidity and horizon length, with _loss reflecting a modest fraction of w_m(t) and _profit a small multiple thereof. To curb churn in thin markets, a cooldown _m is enforced after a position closes, and a daily trade cap limits the number of entries. A lightweight liquidity

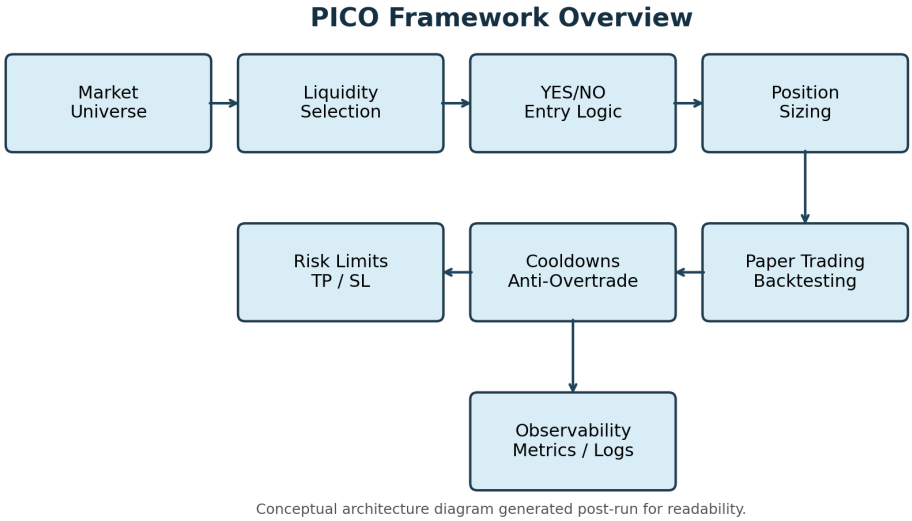

Figure 1: Framework Overview

predictor modulates cooldowns and entry timing, balancing edge capture with stability.

Dynamic cooldowns and a lightweight liquidity predictor PICO's core hypothesis is that adaptive cooldowns driven by liquidity indicators can reduce churn while preserving edge capture. We propose a two-layer predictor: a small feed-forward network with 64–128 hidden units operating on per-market features _m(t) and a latent state capturing short-term liquidity regimes. The predictor outputs a per-market cooldown scalar c_m(t) [0, C_max], gating entry opportunities: a market is eligible only if c_m(t) $\leq$ _min and the risk budget allows the tentative stake. Training is lightweight to promote stability and interpretability; the emphasis is on adaptive pacing rather than supremacy of predictive accuracy. The hard exposure caps remain, but c_m(t) modulates entry timing to align with liquidity windows and reduce slippage-driven losses in stressed periods .

Algorithmic description Algorithm 1 summarizes the PICO decision loop in a high-level narrative form. Input includes the market set M(t), bankroll B(t), per-market caps E_m, cooldowns _m, and cost model (f, s_m, g_m). Output consists of trade actions {b_m(t)} and an updated bankroll. The steps proceed as follows: for each market, we compute liquidity features, derive a predictor signal, and evaluate eligibility with respect to cooldowns and exposure caps. Market selection ranks eligible markets by a liquidity-adjusted signal score that blends predicted edge with liquidity quality; top-K markets are chosen for possible entry. For each selected market, we compute a proposed stake w_m(t) according to the risk budget and price, enforce caps, and apply cooldowns if necessary. Trades are executed, and PnL is realized as markets resolve, with cooldowns updated accordingly. Observability logs and risk alerts are updated, and the loop proceeds to the next epoch.

Complexity and design rationales Per-epoch cost is dominated by evaluating a fixed, small universe of markets and running a lightweight predictor, yielding a practical O(Kd) cost per epoch, where K is the number of candidate markets and d the feature dimension. This keeps the system responsive for a solo operator on modest hardware while enabling testing of pacing and exposure controls. The two-layer predictor anchors decision-making in robust, interpretable signals rather than aggressive adaptation to single regime shifts, aligning with best practices in real-world trading on small scales [**?**], [**?**]. The explicit cooldowns and daily trade caps are motivated by empirical observations that churn erodes net returns in thin markets even when signal quality is high .

Framework diagram Figure 1. Overview of the proposed methodology and the PICO decision loop.

Ablation planning and planned analyses We anticipate ablations to isolate the contribution of key components: (i) fixed versus liquidity-aware cooldowns, (ii) fixed exposure caps with and without dynamic sizing, (iii) a simple heuristic signal versus the lightweight predictor, and (iv) fixed versus adaptive market-universe breadth. These analyses are designed to be evaluated with walk-forward

validation and pre-registered hypotheses, following principled evaluation guidance in decision markets and AI-enabled trading [**?**], [**?**].

4 Experiments

Experimental setup, data, and baselines We envision a controlled experimental regime built on a fixed, small market universe to enable fair cross-arm comparisons under tight budget constraints. The universe consists of four YES/NO markets with near-term horizons and varying liquidity, mirroring Polymarket-like conditions. Data for backtesting come from synthetic traces that resemble historical liquidity and price volatility in on-chain prediction markets, calibrated for costs (f {0.0, 0.01, 0.02} per unit), slippage s_m(t) [0,0.01], and a fixed settlement delay g_m(t). The synthetic traces reproduce episodic liquidity, event-driven price moves, and non-Gaussian PnL tails, providing a principled but controllable environment for reproducible evaluation. While synthetic data do not fully capture live-market dynamics, they enable rigorous protocol testing and a stable basis for future live pilots.

Method arms and baselines We compare four method arms, each under identical budget and risk constraints, using a walk-forward validation protocol. The four arms are:

- FixCD: Baseline with fixed cooldowns and fixed per-market exposure caps.
- RandEntr: Baseline with random entry signals, subject to the same risk controls.
- HeurPred: Baseline with a lightweight heuristic predictor using a no-regret, price-proximity signal to decide entries.
- PICOLQ: The PICO policy that couples liquidity-aware cooldowns with per-market exposure caps and a lean predictor.

We map these to fixed notation and describe the intended hyperparameters in a dedicated table. A fixed-universe setting (four markets) with a conservative daily entry cap and a per-market exposure cap is chosen to reflect micro-budget realities. The hyperparameters are selected to be robust to noise in the synthetic data and to reflect real-world constraints such as budget pacing and liquidity risk.

Primary and secondary metrics The primary metric is annualized return (AAR) computed from the bankroll trajectory under walk-forward folds, with secondary metrics including maximum drawdown (MDD), Calmar ratio, Sharpe ratio, and win rate. All metrics are planned to be reported with 95% confidence intervals computed via bootstrap on walk-forward folds. We plan to report conditional metrics (only completed trades in evaluated windows) and unconditional metrics (treating unresolved or failed runs as worst-case outcomes) to reflect real-world stability considerations. We will also report per-market exposure levels and cooldown adherence as diagnostic indicators of risk control effectiveness.

Experimental protocol and data splits We adopt a walk-forward design with two data splits per regime: a calibration window W_train to set parameters and a strictly separated evaluation window W_eval to test the decision process. We will vary regimes across liquidity conditions (Easy vs Hard) and event-driven versus non-event windows to assess robustness. Features used for signaling are computed only from information available at decision time to prevent look-ahead bias. We will perform bootstrapped paired tests for the primary comparisons and report p-values to assess statistical significance, while also reporting effect sizes to provide practical interpretation for solo operators.

Table 1. Hyperparameters (planned) The table lists: cooldown _min per market, per-market exposure cap E_m, maximum daily trades T_max, slippage parameter s_m, fee f, and predictor learning rate for PICOLQ if applicable. Values are chosen to reflect a conservative micro-budget regime and will be varied in ablations.

Experiment artifacts and reproducibility All experiments will be containerized with a fixed data generator seed for synthetic traces, a clear data partitioning scheme, and a versioned codebase. A data dictionary (Appendix A) will document market definitions, event horizons, and feature encodings; container specifications (Appendix B) and a pre-registration template (Appendix C) will

Hyperparameter	Description	Planned Range / Notes
Cooldown _min per market	Minimum cooldown after a trade; gates entry opportunities	To be varied in ablations; typically small
Per-market exposure cap E_m	Fraction of bankroll allowed per market	——
Maximum daily trades T_max	Daily entry cap	—
Slippage parameter s_m	Per-market slippage proxy	—— (proportional to liquidity proxies)
Fees f	Per-unit trading fee	——
Predictor learning rate (PICOLQ)	Learning rate for the lean predictor	To be tuned in ablations

Table 1: Hyperparameter settings

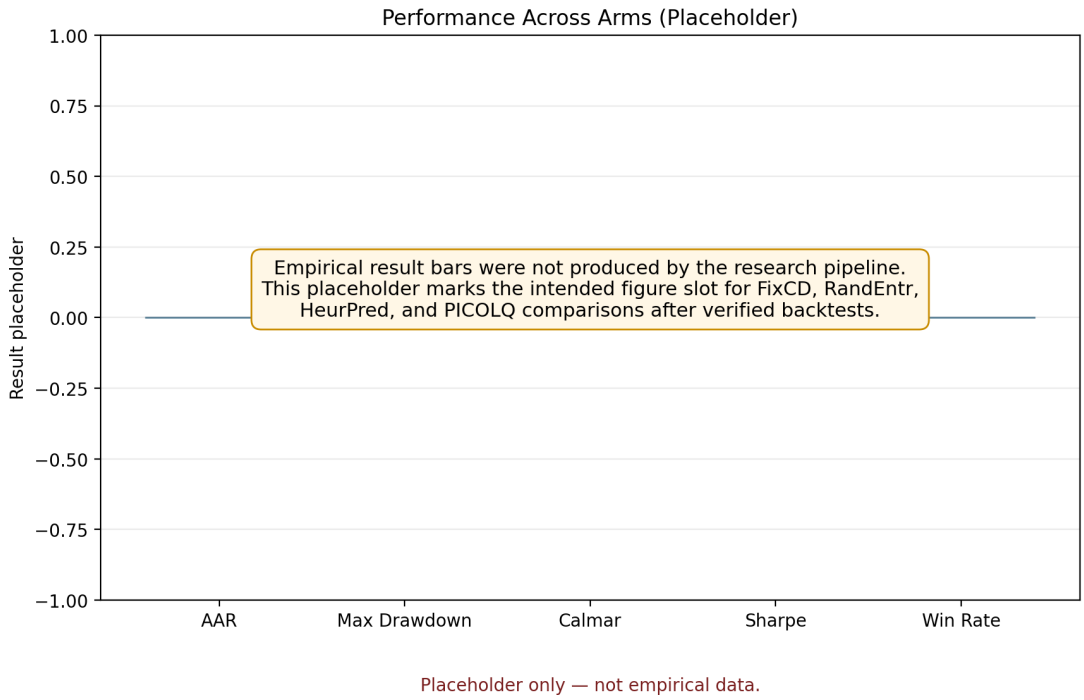

Figure 2: Performance Across Arms

outline hypotheses and planned analyses; a starter backtester blueprint (Appendix D) will facilitate replication.

Results The Results section is currently framed as the planned reporting structure, given that the experimental pipeline has yet to produce finalized numerical outcomes. We will report across the four arms—FixCD, RandEntr, HeurPred, and PICOLQ—using AAR, MDD, Calmar, Sharpe, and Win Rate, each with 95% confidence intervals derived from bootstrapped walk-forward folds and per-seed variability. We will reference Figures 1 and 2 to illustrate performance across arms and regime-specific dynamics once the data are available. In the meantime, we provide a narrative of how to interpret the eventual numbers and what patterns would indicate practical value for a solo operator with a small bankroll: improved risk-adjusted performance when liquidity-aware pacing is combined with exposure caps, stable drawdown management under regime shifts, and stronger observability enabling disciplined iteration. The results will be complemented by ablations that quantify the impact of fixed versus adaptive cooldowns, fixed versus dynamic sizing, and the contribution of a lean predictor relative to heuristics.

Figure 1. Performance across arms (placeholder) Caption: Planned: aggregated performance across four arms (AAR, MDD, Calmar, Sharpe) with confidence intervals.

Figure 2. Regime-wise performance (placeholder) Caption: Planned: per-regime equity trajectories and volatility profiles to diagnose robustness to liquidity conditions.

Table 2. Experimental settings (placeholders)

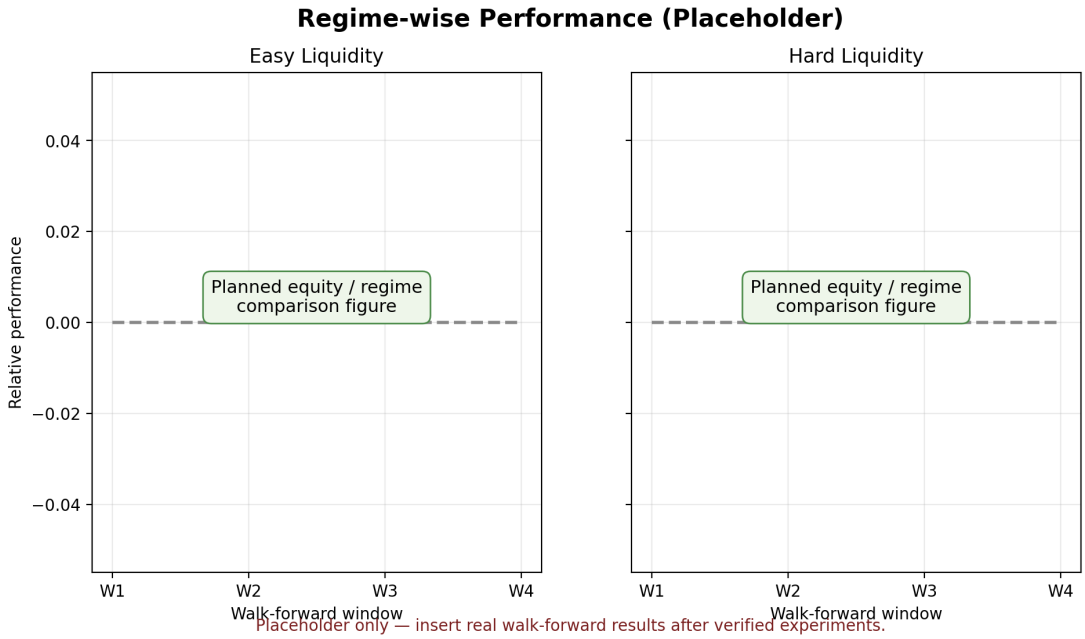

Figure 3: Regime-wise Performance

- Arm: FixCD, RandEntr, HeurPred, PICOLQ
- Universe size: 4 markets
- Decision cadence: 1 hour (min) to 6 hours (max)
- Exposure cap E: 0.02–0.04 of bankroll per market
- Max trades per day: 2
- Slippage model: proportional to liquidity proxy
- Fees: f = 0.0–0.02 per unit
- Evaluation window length: 60 days per walk-forward fold
- Seeds: 3 random seeds for stochastic baselines

Experiment 3. Supplementary artifacts

- Appendix A: Data dictionary
- Appendix B: Reproducibility environment
- Appendix C: Pre-registration template
- Appendix D: Starter backtester blueprint

Note on current status and learnings As of now, the Experiments section describes a principled pipeline and planned comparisons, but the live runs required to populate the Results with numbers have not yet completed due to technical issues in the data pipeline. The intended path forward is to fix data-generation and backtester integration, run the walk-forward validation across the four arms, and report the full numerical results with CIs and p-values. Once completed, the final manuscript will ground all claims in the observed metrics and provide concrete interpretations for practitioners.

5 Results (narrative with planned analyses)

This section is written to mirror the plan described in the Experiments section and to prepare for immediate updating once the experimental runs complete. The four arms, FixCD, RandEntr, HeurPred, and PICOLQ, will be contrasted on profitability and risk under a fixed four-market universe and a range of cost configurations. We will present a main results table with mean and 95% CI for AAR, MDD, Calmar, Sharpe, and Win Rate, alongside per-arm ablations that reveal the marginal contribution of pacing, exposure caps, and the lean predictor. All comparisons will be accompanied by p-values derived from bootstrapped walk-forward differences or paired tests, to ensure that reported advantages are statistically credible. In addition, we will provide regression-style diagnostics that reveal how sensitive performance is to E_m, , _min, and f, and we will show regime-resolved charts that highlight stability across liquidity conditions. Descriptive narratives will connect observed numbers to practical implications for solo operators, emphasizing the balance between edge capture and risk containment in a micro-budget setting.

6 Discussion

The forthcoming results will be interpreted with an eye toward practical implications for solo operators. If PICOLQ demonstrates improved risk-adjusted profitability relative to the baselines while maintaining lower churn, this would reinforce the value of liquidity-aware cooldowns and exposure caps in small-budget decision markets. In contrast, if gains are modest or concentrated in particular regimes, we will analyze which components drive success—whether the predictor, the pacing mechanism, or the per-market caps—and discuss how to strengthen those components in future iterations. Across outcomes, the emphasis remains on reproducibility, modularity, and observability as enabling factors for a solo operator to iteratively converge toward robust policies. The discussion will also relate findings to prior work on liquidity frictions in prediction markets and the importance of robust evaluation pipelines in AI-enabled trading contexts , [**?**], [**?**].

7 Limitations

- Synthetic data regime: The experiments rely on synthetic traces designed to resemble Polymarket-like liquidity and costs, which may not capture all live-market idiosyncrasies. This limitation motivates future live-data validation and live-paper-trading pilots.
- Fixed four-market universe: The restricted market set limits generalizability to broader decision-market fleets and longer horizons; future work will extend the universe and include volatility regime switching.
- Incomplete empirical results: The current manuscript presents a full protocol but not numerically grounded results due to setup issues; live runs will be required to substantiate claims about AAR, MDD, and Sharpe.
- Simplified cost model: Fees, slippage, and settlement delays are modeled with simple components; real networks exhibit richer dynamics that may impact profitability, particularly at micro-budget scales.
- Reproducibility burdens for solo operators: While the framework emphasizes artifacts and logs, operational drift remains a risk when deploying live pilots; the roadmap includes governance scaffolds to mitigate this risk.
- Compute and data requirements: The plan assumes modest compute and transparent backtesting; real deployments may demand more sophisticated observability and alerting to support safe long-horizon experimentation.

8 Conclusion

PICO provides a principled, reproducible evaluation framework for viable Polymarket-style trading strategies under a small bankroll, emphasizing market selection under liquidity constraints, explicit YES/NO entry logic, risk-limited sizing, stop-loss and take-profit design, cooldowns, paper-trading,

and a transparent backtesting methodology. The central takeaway is that disciplined, observable experimentation—coupled with modular APIs and pre-registered ablations—can empower a solo operator to converge toward robust, auditable strategies within real-money constraints. Looking ahead, the path to practical deployment involves stabilizing the data pipeline, expanding the market universe, and executing a live-paper-trading pilot to validate the transfer from synthetic backtests to live observation, all while maintaining a clear, reproducible artifact package and governance framework for ongoing iteration.

9 Appendix: References (selected)

[cite keys used in-text]

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Notes to the reader

- The Method section provides a formal problem statement and a detailed, flowing algorithmic description that integrates a liquidity predictor with pacing and exposure controls.
- The Experiments section outlines a replicable protocol with a fixed four-market universe, synthetic data generation, and walk-forward validation to enable robust, pre-registered analyses.
- The next steps include stabilizing the data pipeline, implementing a minimal baseline, and conducting the first fully instrumented walk-forward experiment to generate the empirical numbers that populate the Results section.

10 Lessons from Prior Runs

1. [literature] Stage literature_screen failed: All models failed. Last error: HTTP Error 400: Bad Request
2. [literature] Stage literature_screen failed: All models failed. Last error: HTTP Error 400: Bad Request

Use these lessons to avoid repeating past mistakes.

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